COURSE ASSIGNMENT SUBMISSION

MediBook: Multi-Agent Clinical POC

FINAL SUBMISSION

Foundations of Applied AI (FAAI) | Academic Year 2026

STUDENT DETAILS

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GITHUB PROJECT REPOSITORY

Click the link below to view the complete source code, agent graphs, and demo:

[Repo] https://github.com/Tanush-o-Kumar/medibook_agentic_poc

PROJECT OVERVIEW & ARCHITECTURE

MediBook is an autonomous multi-agent healthcare coordination and clinical appointment triage system. It implements LangGraph state-driven workflows that coordinate specialized LLM agents for conversational patient intake, urgency assessment, doctor calendar scheduling, and automated notification dispatches. The system is powered by local LLM inference (Ollama Llama 3.2) with a resilient rule-based deterministic fallback.

MULTI-AGENT WORKFLOW MODULES

- **Intake Agent:** Interviews patients, captures medical history, symptoms, and urgency level.
- **Triage & Routing Agent:** Classifies emergency vs routine needs, maps patient to specialized physicians.
- **Scheduling Agent:** Queries physician availability, validates conflicts, and reserves time slots.
- **Notification Agent:** Generates clinical confirmation briefs, patient instructions, and reminder logs.

REPOSITORY FILE STRUCTURE & EXECUTION

app.py	- Streamlit clinical dashboard, chat interface, and doctor schedule views.
graph.py	- LangGraph state machine wiring, routing conditions, and node execution.
agents.py	- Specialized agent implementations, prompt templates, and decision logic.
tools.py	- Synthetic calendar lookup, appointment slot booking, and conflict management.
llm.py	- Ollama client adapter with resilient rule-based deterministic fallback.

To launch the interactive demo: streamlit run app.py